Homework 2

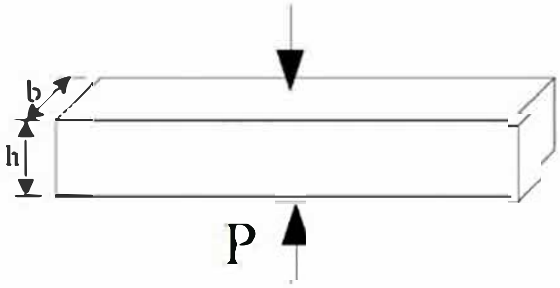
Submission date: 6th May, 2019

1. Prove that $Q[y(x)] = \int_a^b y^2(x)dx$ is not a linear functional.
2. Try to obtain "the shortest curve" between two points by variation method. (beeline)
3. Solve the first order variation of the function $J[y] = \int_0^1 y^3 y'^2 dx$. y(0)=1.
4. Solve the maximum and minimum of the intersection Z between the ellipsoid $16x^2 + 4y^2 + z^2 = 16$ and the plane $x + y + z = 1$.

   Hints: consider the unconditional extremum of the function

   $F = z + \lambda_1(x + y + z - 1) + \lambda_2(16x^2 + 4y^2 + z^2 - 16)$, *where* λ_1 *and* λ_2 *are Lagrange multipliers.*
5. Derive the Euler equation of the extremum condition of the functional containing third-order derivative of its independent function.

6. As shown in the picture apply a pair of holding force P to the middle of a rod with rectangle section. Solve the length change Δ of the rod.



7. Solve the extremum function of the following functional by direct method (the Euler equation) and the indirect method (the Ritz method, select three terms).

$$Q[y]=\int_0^1\left(y'^2-4y^2\right)dx,\quad y(0)=0,\quad y(1)=1$$

8. Solve the Euler equation and the natural boundary condition of the functional below.

$$Q[y]=\frac{1}{2}\int_{x_0}^{x_1}\left[p_{(x)}.y''^2+q_{(x)}.y'^2+r_{(x)}.y^2-2s_{(x)}.y\right]dx$$